

t-SNE Problem Boot Camp

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1 Summary and Overview

This section is used to summarize the main idea and process of the paper. This is a good process for learning and it would be beneficial to check with Tuca whether the understanding of the big picture is correct. Later in this section we will highlight the main difference between this document and the original paper should the readers are looking for the lemmas associated with the general output kernel.

Difference from the Original Paper

2 Progress and Roadmap for Ph.D Thesis

This section is used to lay out the plan for next steps and how we could finish the Ph.D thesis on time.

3 Setup

Here goes the real math. Recall that this document is dedicated to the reconstruction of [1] with low-dimension kernel g replaced by a more general kernel. This would serve as a great way to understand the material and help with the completion of the original project. For the sake of simplicity, we will use the exact same notation as in [1] and skip all the definitions. Recall in [1], g is defined as

$$g(y, y') = \frac{1}{1 + \|y - y'\|^2} \quad (3.1)$$

With all these being said, we will first start by giving our general low dimension kernel g some general assumptions, and modify it as we move on. We first require that $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (3.2)$$

We will kick off this warmup by listing all the important lemmas and results in order, and prove our own versions when the need arises. The final goal is to prove a same set of results as in [1] with the y kernel replaced. Recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) = & - \int \frac{\|x - x'\|^2}{2\sigma^2} \frac{\exp(-\|x - x'\|^2 / 2\sigma^2)}{\int \exp(-\|x - x'\|^2 / 2\sigma^2) d\mu(x')} d\mu(x') \\ & - \log \left(\int \exp(-\|x - x'\|^2 / 2\sigma^2) \right) + \log \rho. \end{aligned} \quad (3.3)$$

Proposition 1. *If μ has a C^1 density f such that both f and $\|\nabla f\|$ are bounded, then the equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. This proposition is about the kernel of x . The proof stays unchanged from [1]. □

The next couple of lemmas are general results in section 3 of [1]. We will omit the proof because they are not directly discussing the quantities in the t-sne algorithm. Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

Lemma 1. *Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that*

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq C e^{-c_1 t^2}. \quad (3.4)$$

Lemma 2. *Let μ be sub-Gaussian and c_1 be the constant from lemma 1. Let $X_1, \dots, X_n \in \mathbb{R}^d$ have i.i.d. entries drawn from μ . Then there exists a constant C such that*

$$\mathbb{P} \left(\max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\frac{\log n}{c_1}} + \sqrt{d} + t \right) \leq C e^{-c_1 t^2}. \quad (3.5)$$

Lemma 3. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$.
Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (3.6)$$

Define

$$f_\sigma(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2), \quad g_\sigma = \frac{\|x - x'\|^2}{2\sigma^2} \exp(-\|x - x'\|^2 / 2\sigma^2). \quad (3.7)$$

The next two lemmas use similar techniques to prove.

Lemma 4. Let c_1 be the decay constant in lemma 1 and let $\delta > 0$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu_n(x')} - \frac{\int g_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| = O(n^{8/c_1 \delta^2 - 1/2} \sqrt{\log n}). \quad (3.8)$$

Lemma 5. Let c_1 be the decay constant in lemma 1 and let $\delta > 0$ be such that $c_1 \delta^2 > 8$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \log \left(\int f_\sigma(X_i, x') d\mu_n(x') \right) - \log \left(\int f_\sigma(X_i, x') d\mu(x') \right) \right| = O(n^{4/c_1 \delta^2 - 1/2} \sqrt{\log n}) \quad (3.9)$$

For this pair of lemmas, [1] provides the proof to the first lemma. We will provide the second for the sake of exercise.

Proof of lemma 5. First, notice that for arbitrary $x > 0$, the following elementary inequality holds:

$$\log(x) \leq x - 1. \quad (3.10)$$

Therefore, the lemma can be proved if we are able to show that

$$\begin{aligned} \left| \frac{\int f_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu(x')} - 1 \right| &= \left| \frac{\int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \\ &= O(n^{4/c_1 \delta^2 - 1/2} \sqrt{\log n}). \end{aligned} \quad (3.11)$$

We can achieve that by bounding the numerator and the denominator of the previous formula. On one hand, consider set

$$E'_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\alpha + \frac{1}{c_1} \log n} + \sqrt{d} \right\} \quad (3.12)$$

Setting $t = \sqrt{\alpha \log n}$ in lemma 2, we have $\mathbb{P}(E'_1) \leq Cn^{-c_1 \alpha}$. To absorb the $\sqrt{d}$ term, we can find $\epsilon_n \rightarrow 0$ such that

$$E_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\alpha + \frac{1}{c_1} \log n} + \epsilon_n \sqrt{\log n} \right\} \quad (3.13)$$

with $\mathbb{P}(E_1) \leq Cn^{-c_1\alpha}$. Because the maximum of norm of each X_i is restricted, on E_1^c we have for all i, j , that

$$\|X_i - X_j\|^2 \leq 4 \left(\alpha + \frac{1}{c_1} + \epsilon_n \right) \log n =: 4\gamma \log n. \quad (3.14)$$

Then for $\sigma \in [\delta, \sqrt{n}]$, on E_1^c for all i we have

$$\int f_\sigma(X_i, x') d\mu(x') \geq (1 - Cn^{-c_1\alpha}) \exp\left(-\frac{2\gamma \log n}{\delta^2}\right) \geq \frac{1}{2} \exp\left(-\frac{2\gamma \log n}{\delta^2}\right) = \frac{1}{2} n^{-2\gamma/\delta^2}. \quad (3.15)$$

This helps us bound the denominator. On the other hand, by lemma 3, we are able to find set E_2 with $\mathbb{P}(E_2) \leq e^{-c\epsilon^2 n}$, such that, on E_2^c ,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| \leq \epsilon + O\left(\frac{\sqrt{\log n}}{\sqrt{n}}\right). \quad (3.16)$$

Therefore, we can combine (3.15) and (3.16) and conclude that on $E_1^c \cap E_2^c$,

$$\left| \frac{\int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \leq C\epsilon n^{2\gamma/\delta^2} + O(\sqrt{\log n} \cdot n^{-1/2+2\gamma/\delta^2}). \quad (3.17)$$

In order for the RHS to converge to zero, we need

$$\epsilon n^{2\gamma/\delta^2} \rightarrow 0, \quad 1/2 - 2\gamma/\delta^2 > 0. \quad (3.18)$$

Moreover, we have $\mathbb{P}(E_1 \cup E_2) \leq Cn^{-c_1\alpha} + e^{-c\epsilon^2 n}$. Set $\epsilon = n^{-\eta}$ for some $\eta > 0$. In order to get almost sure convergence using Borel-Cantelli, we also need

$$c_1\alpha > 1, \quad 1 - 2\eta > 0. \quad (3.19)$$

Hence for the entire lemma to work we need the following four inequalities to hold:

$$\frac{2\gamma}{\delta^2} - \eta < 0, \quad \frac{2\gamma}{\delta^2} - 1/2 < 0, \quad c_1\alpha > 1, \quad 1 - 2\eta > 0. \quad (3.20)$$

The last inequality implies that $\eta < 1/2$ so the first inequality implies the second. Now, recall that

$$\gamma := \alpha + \frac{1}{c_1} + \epsilon_n. \quad (3.21)$$

The collection of inequalities is equivalent to

$$2 \left(\alpha + \frac{1}{c_1} + \epsilon_n \right) < \eta \delta^2, \quad c_1\alpha > 1, 0 < \eta < 1/2. \quad (3.22)$$

Here, we are free to choose α and η , but c_1 and δ are fixed. Now let

$$\alpha = \frac{1}{c_1} + \epsilon, \quad \eta = \frac{1}{2} - \epsilon. \quad (3.23)$$

Then the second and third inequalities in (3.22) are satisfied. The first is equivalent to

$$\frac{4}{c_1\delta^2} + \epsilon < \frac{1}{2} - \epsilon \quad (3.24)$$

as $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$. In addition, since ϵ is arbitrary, we have again, equivalently,

$$\frac{4}{c_1 \delta^2} - \frac{1}{2} < 0. \quad (3.25)$$

When this is satisfied, we have:

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| = O(n^{4/c_1 \delta^2 - 1/2} \sqrt{\log n}). \quad (3.26)$$

The proof is thus finished. $\square$

Then combining the previous result we have an important convergence:

Proposition 2. *Let μ be sub-Gaussian and c_1 be the constant from lemma 1. $X_1, \dots, X_n$ are i.i.d. random variables with probability measure μ and $\mu_n = \frac{1}{n} \sum \delta_{X_i}$ is the empirical measure. Additionally, set $\delta > 0$ to be a constant such that $c_1 \delta^2 > 16$. Then, there exists $\epsilon > 0$, such that almost surely in $X_1, \dots, X_n$:*

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} |F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)| = O(n^{-\epsilon}). \quad (3.27)$$

Let us summarize the important results upon this point, which will be used later:

There exists $\gamma = 2/c_1 + \epsilon$ such that the following holds almost surely

- (1) $\max_{i=1, \dots, n} \|X_i\| \leq \sqrt{\gamma \log n}$;
- (2) $\max_{i,j} \|X_i - X_j\|^2 \leq 4\gamma \log n$;
- (3) For $\sigma \in [\delta, \sqrt{n}]$,

$$\int \exp(-\|X_i - x'\|^2 / 2\sigma^2) d\mu_n(x), \int \exp(-\|X_i - x'\|^2 / 2\sigma^2) d\mu(x') \geq C n^{-2\gamma/\sigma^2}. \quad (3.28)$$

Next couple of lemmas in [1] are about lower bound on $\sigma_{\rho, \mu}^*$. Recall that $\sigma_{\rho, \mu}^*$ is the unique solution of the equation $F_{\rho, \mu}(x, \sigma_{\rho, \mu}^*) = 0$. For simplicity of notation, we write $\sigma^*(x) = \sigma_{\rho, \mu}^*(x)$ and $\sigma_n^*(x) = \sigma_{\rho, \mu_n}^*(x)$.

Lemma 6. *If μ satisfies the condition in lemma 1 of [1], then $\sigma^*(x)$ and $\sigma_n^*(x)$ are uniformly bounded from below in both i and n almost surely.*

Proposition 3. *Suppose μ has compact support and C^1 density. Then almost surely, as $n \rightarrow \infty$,*

$$\sup_{i=1, \dots, n} |\sigma_n^*(X_i) - \sigma^*(X_i)| \rightarrow 0. \quad (3.29)$$

Proof. By [1], we can quickly reach to the stage where

$$|\sigma_n^*(x) - \sigma^*(x)| = O(|F_{\mu_n}(x, \sigma_n^*(x)) - F_\mu(x, \sigma_n^*(x))|) \rightarrow 0. \quad (3.30)$$

Moreover, we know that μ has compact support and $x \mapsto \sigma^*(x)$ is smooth, $\sigma^*(X_i)$ is bounded above and below, meaning that $\sigma_n^*(X_i)$ is bounded uniformly from above and below as well. This completes the proof in the original paper. $\square$

4 More Lemmas

Starting this section we are gradually approaching our point of work. We need to list a few more lemmas before that. To make notations cleaner, [1] introduced some abbreviation in section 4.2. We will use all of them by default.

Lemma 7. *Any sequence in $\tilde{\mathcal{P}}_X$ is tight.*

Recall Prokhorov's theorem:

Theorem (Prokhorov's Theorem). *If μ_n is a tight sequence of measures in Euclidean space. There exists a subsequence μ_{n_k} and a measure μ such that μ_{n_k} converges to μ weakly.*

We immediately get the following lemma:

Lemma 8. *There exists μ^* such that*

$$I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu) \quad (4.1)$$

Recall that $I_\rho(\mu)$ is the continuous version of $L_{n,\rho}(X, Y)$. It only depends on the measure μ and the perplexity constant ρ . We have not reached the part that requires we have a common perplexity, but from now on we should pay close attention as it is one of the core setup of the original paper.

Lemma 9. *As $n \rightarrow \infty$, for some $\epsilon > 0$, we have*

$$|f_n(X_i, X_j) - f(X_i, X_j)| = O(n^{-\epsilon}). \quad (4.2)$$

Proof. There are typos in the original proof so we prove this lemmas again despite it being very easy and elementary. We compute directly

$$\begin{aligned} & |f_n(x, x') - f(x, x')| = |\exp(-\|x - x'\|^2 / 2\sigma_n(x)^2) - \exp(-\|x - x'\|^2 / 2\sigma(x)^2)| \\ &= \left| \exp\left(\frac{|\sigma_n(x)^2 - \sigma(x)^2| \|x - x'\|^2}{2\sigma_n(x)^2 \sigma(x)^2}\right) - 1 \right| \cdot \exp(-\|x - x'\|^2 / 2\sigma(x)^2) \\ &\leq \left| \exp\left(\frac{|\sigma_n(x)^2 - \sigma(x)^2| \|x - x'\|^2}{2\sigma_n(x)^2 \sigma(x)^2}\right) - 1 \right|. \end{aligned} \quad (4.3)$$

Recall that the sum of two sub-Gaussian random variables is still sub-Gaussian. If μ_X is sub-Gaussian, let C be the constant of decay for $X_i - X_j$ and we have

$$\mathbb{P}(\|X_i - X_j\|^2 \geq \log n) \leq 2 \exp(-\log n / C^2) = 2 \exp\left(-\frac{1}{C^2}\right) \cdot \frac{1}{n} \rightarrow 0. \quad (4.4)$$

This means $\|X_i - X_j\|^2 = O(\log n)$ almost surely. On the other hand, if μ_X has compact support then $\|X_i - X_j\|$ will be bounded, even easier. Moreover, since σ_n and σ are both uniformly bounded below by lemma (6), by setting $x = X_i$ and $x' = X_j$ we get

$$\begin{aligned} & \left| \exp\left(\frac{|\sigma_n(x)^2 - \sigma(x)^2| \|x - x'\|^2}{2\sigma_n(x)^2 \sigma(x)^2}\right) - 1 \right| \\ &\leq \frac{1}{\sigma_n(x)\sigma(x)} \left(\frac{1}{\sigma_n(x)} + \frac{1}{\sigma(x)} \right) |\sigma_n(x) - \sigma(x)| \|x - x'\|^2 \\ &= O(|\sigma_n(x) - \sigma(x)| \|x - x'\|^2) \\ &= O(|\sigma_n(x) - \sigma(x)|) O(\log n) \\ &= O(n^{-\epsilon}) O(\log n) \\ &= O(n^{-\epsilon}), \end{aligned} \quad (4.5)$$

by proposition (3). This finishes the proof. $\square$

One thing we should highlight here is that we are abusing the notation of σ and σ_n . They are defined as the unique solution to $F = 0$. However, the definition of σ_i is not abused and we will later see why such abuse makes sense.

Lemma 10. *If $c_1\delta > 32$, then there exists $\epsilon > 0$ such that as $n \rightarrow \infty$, we have*

$$\begin{aligned} \sup_{i,j} |p'_n(X_i, X_j) - p'(X_i, X_j)| &= O(n^{-\epsilon}), \\ \sup_{i,j} |p_n(X_i, X_j) - p(X_i, X_j)| &= O(n^{-\epsilon}). \end{aligned} \quad (4.6)$$

Supplementary Proof. The overall methodology of the original proof in [1] is natural and clear. However, it can still be slightly confusing as it is missing the actual values we assign to ϵ . In the original paper, we can take, for some constant $k > 0$, that

$$n^{-\epsilon} = n^{8/c_1\delta^2 - 1/2} (\log n)^k. \quad (4.7)$$

This comes from the detailed computations in the original proof and the quantity above is the slowest decaying speed, the 'worst case scenario' given all the conditions. Plug this into the computations and everything is clear. The proof uses a little bit extra than just the bounds in section 3.1. $\square$

Lemma 11. *If $c_1\delta > 40$, then there exists $\epsilon > 0$ such that as $n \rightarrow \infty$, we have*

$$\sup_{i,j} |p_n(X_i, X_j) \log p_n(X_i, X_j) - p(X_i, X_j) \log p(X_i, X_j)| = O(n^{-\epsilon}). \quad (4.8)$$

Lemma 12. *If $c_1\delta^2 > 8$, then as $n \rightarrow \infty$ we have*

$$\sum_j p_{j|i} \log p_{j|i} + \log(\rho n) = o(1). \quad (4.9)$$

[Yi: TODO: Ask Tuca about the last step of the proof in [1].]

5 The Actual Computation

From this point onward, we are starting to have lemmas and results related to the output kernel g and this is where the trail begins. Along the way we will see if there is a need to add more conditions to g and if we do, every time we add more constraints we will reiterate the entire settings for g for clarity. To refresh our memory, here is the initial assumptions we have for g : $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0,\infty)^2} k(x, x') dx dx' < \infty. \quad (5.1)$$

Lemma 13. *There exists a set of points $y^*(X)$ such that*

$$\inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y) = L_{n,\rho}(X, y^*). \quad (5.2)$$

[Yi: TODO: Ask Tuca about the $1/n^2$ in the original proof. The scaling seems to not play a role in the proof.]

Proof. Fortunately, the first proof is almost identical to the original one. Let $k \in \{1, \dots, n\}$. We are going to show that

$$\lim_{|y_k| \rightarrow \infty} L_{n,\rho}(X, y) = \infty. \quad (5.3)$$

By definition, L is the relative entropy and

$$L_{n,\rho}(X, y) = \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (5.4)$$

For n given and fixed, it suffices to show that

$$\sum_{i \neq j} p_{ij} \log(q_{ij}) = \sum_{i \neq j} p_{ij} \log \left(\frac{g(y_i, y_j)}{\sum_{m \neq l} g(y_m, y_l)} \right) \rightarrow -\infty. \quad (5.5)$$

Recall that $g(y, y')$ is bounded. We have

$$\begin{aligned} & - \sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq l} g(y_m, y_l) \right) = - \log \left(\sum_{m \neq l} g(y_m, y_l) \right) \\ & \leq - \sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq k, l \neq k, m \neq l} g(y_m, y_l) \right) < \infty. \end{aligned} \quad (5.6)$$

On the other hand, we have $g(y_i, y_k) = k(\|y_i - y_k\|) \rightarrow 0$ as $|y_k| \rightarrow \infty$ because $\|y_i - y_k\| \rightarrow \infty$ and k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (5.7)$$

This means

$$\lim_{|y_k| \rightarrow \infty} \sum_{i \neq j} p_{ij} \log(g(y_i, y_j)) \rightarrow -\infty. \quad (5.8)$$

Hence, the minimum of the $L_{n,\rho}(X, y)$ is achieved on a bounded set, the lemma follows by the continuity of $L_{n,\rho}(X, y)$. $\square$

Lemma 14. *As $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} p_{ij} \log p_{ij} = \iint p(x, x') \log p(x, x') d\mu d\mu - 2 \log n + o(1). \quad (5.9)$$

Proof. [Yi: TODO: In the proof we did not use lemma 7 of the original paper. Ask Tuca what did I miss.] The original paper does provide a detailed proof, so it is our turn to exercise. First recall that $p_{ij} = p_n(X_i, X_j)/n^2$. Therefore we have

$$\begin{aligned} \sum_{i \neq j} p_{ij} \log p_{ij} &= \sum_{i \neq j} \frac{1}{n^2} p_n(X_i, X_j) \log \left(\frac{1}{n^2} p_n(X_i, X_j) \right) \\ &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - 2 \log n \sum_{i \neq j} \frac{1}{n^2} p_n(X_i, X_j) \\ &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - 2 \log n, \end{aligned} \quad (5.10)$$

because the second summation is just the sum of all probability mass. Now we have the $2\log n$ term and we are left to deal with the first summation. Inspired by the proofs in [1], we begin by showing that the diagonal terms do not matter in our estimation. consider

$$\begin{aligned}
& \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - \iint p_n(x, x') \log p_n(x, x') d\mu_n(x) d\mu_n(x') \\
&= -\frac{1}{n^2} \sum_{t=1}^n p'_n(X_t, X_t) \\
&= -\frac{1}{n^2} \sum_{t=1}^n \frac{1}{\int f(X_t, x') d\mu_n(x') - \frac{1}{n}} \log \left(\frac{1}{\int f(X_t, x') d\mu_n(x') - \frac{1}{n}} \right). \tag{5.11}
\end{aligned}$$

If we refer to the proof of lemma 14 in [1], we can see that the difference of diagonal term vanishes and it does not depend on the specific X_i . Therefore in our proof, when we sum over all the X_i and normalize by dividing n^2 , we reach the exact same conclusion. The diagonal term is $O(n^{4/c_1\delta^2-1} \log n)$ which is $o(1)$ depending on our choice of c_1 .

Then we apply lemma 11 and since $d\mu_n \times d\mu_n$ is a probability measure, we have

$$\iint p_n(x, x') \log p_n(x, x') d\mu_n(x) d\mu_n(x') = \iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') + o(1), \tag{5.12}$$

which immediately implies

$$\sum_{i \neq j} p_{ij} \log ij = \iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') - 2\log n + o(1). \tag{5.13}$$

The proof will be finished if we are able to replace the measure in the last integral with $d\mu$. We verify that

$$h(x, x', \sigma) := p(x, x') \log p(x, x') \tag{5.14}$$

satisfies all the requirements in lemma 3. First μ is sub-Gaussian by construction. Then is easy to verify that h is C^1 for all x, x' and σ . Moreover, by calculus we have $p(x, x') \in (0, 1]$ and $\lim_{a \rightarrow 0} a \log a = 0$. Thus $p \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$. It is also obvious that h is bounded and $h \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$. The rest of the conditions are also very easily to prove. We can finally conclude that

$$\iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') = \iint p(x, x') \log p(x, x') d\mu(x) d\mu(x') + o(1). \tag{5.15}$$

Combine this estimation with (5.13) and we can conclude

$$\sum_{i \neq j} p_{ij} \log ij = \iint p(x, x') \log p(x, x') d\mu(x) d\mu(x') - 2\log n + o(1), \tag{5.16}$$

as desired. □

Partial Derivative of the Relative Entropy

Before we move on, we would like to take a small deviation and compute the partial derivative of the relative entropy. This is important because it will impact the property of the set $\hat{\mathcal{P}}_X$.

Lemma 15. Let $X \in (\mathbb{R}^d)^n$ and $Y = (y_1, \dots, y_n) \in (\mathbb{R}^s)^n$ be the input and output of the t -SNE algorithm, then we have

$$\frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \quad (5.17)$$

Proof. By definition,

$$\begin{aligned} L_{n,\rho}(X, Y) &= \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \\ &= \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \end{aligned} \quad (5.18)$$

The first summation does not depend on y_i and will vanish when taking the partial derivative. Moreover, we can further breakdown the second summation:

$$\begin{aligned} - \sum_{i \neq j} p_{ij} \log q_{ij} &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \sum_{i \neq j} p_{ij} \log \left(\sum_{k \neq l} g(y_k, y_l) \right) \\ &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \log \left(\sum_{k \neq l} g(y_k, y_l) \right). \end{aligned} \quad (5.19)$$

Taking derivative of the last expression and note that only terms that contain y_i remains. This means

$$\begin{aligned} \frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{1}{\sum_{k \neq l} g(y_k, y_l)} \frac{\partial}{\partial y_i} g(y_i, y_j) \right) \\ &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{g(y_i, y_j)}{\sum_{k \neq l} g(y_k, y_l)} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} \right) \\ &= - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \end{aligned} \quad (5.20)$$

□

We can easily do a sanity test by letting $g(y_i, y_j) = (1 + \|y_i - y_j\|^2)^{-1}$ and get formula (A.1) in [1].

Back to the Main Arena

The derivative in lemma 15 explains why $\hat{\mathcal{P}}_X$ has that specific form. We are gradually marching towards the end. Onto the next lemma we go:

Lemma 16. The following holds:

$$\liminf_{n \rightarrow \infty} \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \geq \iint p(x, x') \log \frac{p(x, x')}{q(y, y')} d\mu d\mu. \quad (5.21)$$

Proof. Good news, the proof of this lemma also requires minimal details of the output kernel and is just a simple application of the Fatou's lemma. Like in lemma 15, we first break down

the summation:

$$\sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (5.22)$$

The convergence of the first summation has been proved in lemma 14. We just need to deal with the second summation. Note that we now have

$$\mu_n = \frac{1}{n} \sum \delta_{X_i} \times \delta_{y_i} \quad (5.23)$$

if y_i is involved. By the definition of q_{ij} , we have

$$\begin{aligned} & - \sum_{i \neq j} p_{ij} \log(q_{ij}) \\ &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log g(y_i, y_j) + \log \left(\frac{1}{n^2} \sum_{i \neq j} g(y_i, y_j) \right) + 2 \log n \\ &= \iint p_n(x, x') g(y, y') d\mu_n d\mu_n + \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) + 2 \log n. \end{aligned} \quad (5.24)$$

By Fatou's lemma and lemma 10, we have

$$\liminf_{n \rightarrow \infty} \iint p_n(x, x') \log g(y, y') d\mu_n d\mu_n \geq \iint p(x, x') \log g(y, y') d\mu d\mu. \quad (5.25)$$

Similarly, using Fatou lemma again and we have

$$\liminf_{n \rightarrow \infty} \iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \iint g(y, y') d\mu d\mu. \quad (5.26)$$

The $1/n$ is the duplicated diagonal term in the empirical measure and will obviously vanish as we take $n \rightarrow \infty$. Moreover, because of our assumption on the kernel g , the integrals on the RHS of the last two inequalities converge. In addition, since $\log$ is a continuous and increasing function, (5.26) implies

$$\liminf_{n \rightarrow \infty} \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) \geq \log \left(\iint g(y, y') d\mu d\mu \right). \quad (5.27)$$

The proof is finished by combining (5.24), (5.25), (5.27) and lemma 14. The $2 \log n$ cancelled out because of (5.13). □

In the original paper, this is the point where we finish proving the lower bound of the main theorem. However, we need to postpone such triumph later as we need a later lemma to reconstruct the results in the appendix of [1]. For convenience, define

$$q_n(y, y') = \frac{g(y, y')}{\iint g(y, y') d\mu_n(y) d\mu_n(y') - \frac{1}{n}}. \quad (5.28)$$

By lemma 8 there exists mu^* such that $I(\mu^*) = \inf_{\mu} I_{\rho}(\mu)$. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. random variables drawn from μ^* and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{(X_i, Y_i)}$ their empirical measure. μ_n converges to μ^* weakly.

Lemma 17. *As $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \iint p_n(x, x') \log \left(\frac{p_n(x, x')}{q_n(y, y')} \right) d\mu_n d\mu_n + o(1). \quad (5.29)$$

Proof. We begin with the familiar computation

$$\begin{aligned} \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log \left(\frac{p_n(X_i, X_j)}{q_n(X_i, X_j)} \right) \\ &= \iint p_n(x, x') \log \left(\frac{p_n(X_i, X_j)}{q_n(X_i, X_j)} \right) d\mu_n d\mu_n + \text{Err}_n, \end{aligned} \quad (5.30)$$

where

$$\begin{aligned} \text{Err}_n &= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \left(\log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) - \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right) \right) \\ &= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) \\ &\quad + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \end{aligned} \quad (5.31)$$

[Yi: Ask Tuca about the errors in the original paper.] By (4.4), we know that $\|X_i - X_j\| \leq \log n$ almost surely as $n \rightarrow \infty$. This means $f_n(X_i, X_k) \geq 1/n + \exp(-2\sigma_n)$ almost surely. Using this fact it is easy to see that the first summation in the error term is $o(1)$ and we can write

$$\text{Err}_n = o(1) + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \quad (5.32)$$

For the second term, it is already clear that

$$\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} = o(1). \quad (5.33)$$

On the other hand, we have by Fatou's lemma:

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k \neq l} g(Y_k, Y_l) = \liminf_{n \rightarrow \infty} \iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \iint g(y, y') d\mu^* d\mu^* < \infty, \quad (5.34)$$

by our assumption on the output kernel g . We can see that the second term in Err_n is also $o(1)$, which immediately implies $\text{Err}_n = o(1)$ and completes the proof. $\square$

The next lemma is needed to closing up the lower bound part of the main theorem.

Lemma 18. *As $n \rightarrow \infty$, we have*

$$\iint g(y, y') d\mu_n(y) d\mu_n(y) - \frac{1}{n} = \iint g(y, y') d\mu^* d\mu^* + o(1). \quad (5.35)$$

Proof. In this proof we will be adding some minor constraints to $g(y, y')$ to make the proof work. Since $\mu_n \rightarrow \mu$ weakly, it is tight by Prokhorov's theorem. For arbitrary $\epsilon > 0$, there exists a compact set K such that for all n , $\mu^*(K^c), \mu_n(K^c) \leq \epsilon$. **[Yi: TODO: Ask Tuca**

about the typo in the original paper.] Since $g(y, y')$ is bounded, there exists a constant C such that

$$\begin{aligned} & \left| \iint g(y, y') d\mu_n(y) d\mu_n(y) - \iint g(y, y') d\mu^* d\mu^* \right| \\ &= \left| \iint g(y, y') \mathbb{1}_{K \times K} (d\mu_n d\mu_n - d\mu^* d\mu^*) \right| + C\epsilon. \end{aligned} \quad (5.36)$$

Define

$$G_n(y) = \int g(y, y') \mathbb{1}_K d\mu_n(y'), \quad G(y) = \int g(y, y') \mathbb{1}_K d\mu^*(y'). \quad (5.37)$$

We now require the kernel g to have continuous and bounded derivative, and we have the following claims:

Claim 1. $G_n(y)$ is uniformly equicontinuous on K .

Proof of Claim. Recall that by our assumptions $g(y, y') = k(\|y - y'\|)$. We get

$$\begin{aligned} |G_n(y) - G_n(\tilde{y})| &= \left| \int k(\|y - y'\|) - k(\|\tilde{y} - y'\|) \mathbb{1}_K d\mu_n(y') \right| \\ &\leq \int \|y - \tilde{y}\| \left| \frac{k(\|y - y'\|) - k(\|\tilde{y} - y'\|)}{\|(y - y') - (\tilde{y} - y')\|} \right| \mathbb{1}_K d\mu_n(y') \\ &= O(\|y - \tilde{y}\|). \end{aligned} \quad (5.38)$$

by mean value theorem and the additional assumption we just added. $\square$

Claim 2. We have $\sup_{y \in K} |G_n(y) - G(y)| = o(1)$.

Proof of Claim. Since K is compact, it suffices to show that for each $y \in K$, there exists $\delta > 0$ such that

$$\sup_{\tilde{y} \in B(y, \delta)} |G_n(\tilde{y}) - G(\tilde{y})| \leq \epsilon. \quad (5.39)$$

[Yi: TODO: Ask Tuca how exactly compactness plays a role in the proof. Using the finite subcover of all the balls?] Towards this end, by triangular inequality,

$$|G_n(\tilde{y}) - G(\tilde{y})| \leq |G_n(\tilde{y}) - G_n(y)| + |G_n(y) - G(y)| + |G(y) - G(\tilde{y})|. \quad (5.40)$$

By the previous claim, the first term on the RHS vanishes because of equicontinuity. The second term vanishes because of weak convergence and the boundedness of the integrand $g(y, y')$. Finally, the last term vanishes because we can prove the equicontinuity of G using a similar argument like the claim 1. $\square$

By claim 2 we have

$$\iint g(y, y') \mathbb{1}_{K \times K} d\mu_n d\mu_n = \iint g(y, y') \mathbb{1}_{K \times K} d\mu^* d\mu^* + o(1) = \int G(y) \mathbb{1}_K d\mu_n(y) + o(1). \quad (5.41)$$

Since G is bounded and continuous by our assumptions on the kernel $g(y, y')$, using weak convergence of μ_n and we have

$$\int G(y) \mathbb{1}_K d\mu_n(y) = \int G(y) \mathbb{1}_K d\mu^*(y) + o(1) = \iint g(y, y') d\mu^* d\mu^* + o(1), \quad (5.42)$$

as desired. $\square$

For clarify, we reiterate our assumptions: The output kernel satisfies $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (5.43)$$

3. k has continuous and bounded derivative.

We are near the end, a couple of more lemmas to go!

Lemma 19. *If μ^* has compact support in Y , then*

$$\iint p_n(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p_n(x, x') \log(g(y, y')) d\mu^* d\mu^* + o(1). \quad (5.44)$$

Lemma 20. *If μ^* has compact support in Y , then*

$$\sum_{i \neq j} p_{ij} \log\left(\frac{p_{ij}}{q_{ij}}\right) = \iint p(x, x') \log\left(\frac{p(x, x')}{q(y, y')}\right) d\mu^* d\mu^* + o(1) \quad (5.45)$$

Proposition 4. *If μ^* has compact support in Y , then*

$$\limsup_{n \rightarrow \infty} \inf_{Y \in (\mathbb{R}^s)^n} L_{n, \rho}(X, Y) \leq I_\rho(\mu^*). \quad (5.46)$$

6 Finishing up the Lower Bound

7 Proof of the Main Theorems

References

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